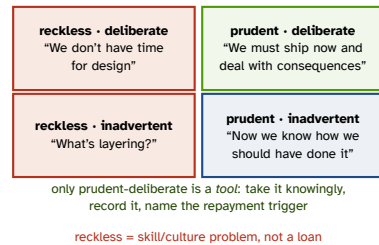
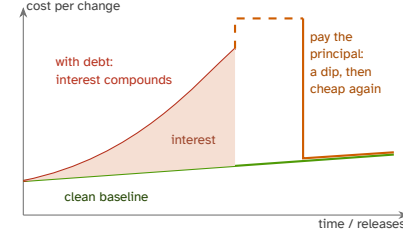


In one line: Technical debt (Cunningham, 1992) is the extra cost future changes pay because of a shortcut taken — or a better design learned — earlier. **Principal** = the effort to fix it; **interest** = the extra effort (and risk) every change in that area pays until you do. Code nobody touches charges no interest. Manage debt like a portfolio: make it visible, price it in product terms, pay where interest is highest — and record *why* big decisions were made (**ADRs**) so nobody re-litigates them in a year.

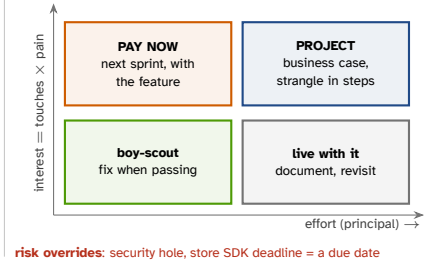
Fowler's debt quadrant (2009)



Principal vs interest



What to pay first



How it works

- Kinds:** code (duplication, god objects) · architecture (wrong boundaries, a shared singleton everything reaches) · test (no seams, flaky suite) · **dependency** (old SDKs, deprecated APIs, an abandoned library — on mobile the *stores* set the due date) · build/CI (slow, manual release steps) · knowledge (one person understands payments: bus factor 1).
- Measure the interest,** not the ugliness: lead time for changes in that area, bugs and incidents per module, change-failure rate, **hotspots** (churn × complexity), build and CI time, flaky-test rate, share of unplanned work, dependency age. Before/after numbers make the case.
- Prioritise:** interest (how often the code is touched × how much each touch hurts) × risk (security, crash, deadline) / effort. Pay *adjacent* to roadmap work ("to build X we first split Y"); keep a standing share of capacity for the rest; big items become projects with milestones, each one shippable.
- Register it:** a ticket per debt item with the interest observed, the principal estimate and the *trigger* ("when we add a 2nd payment provider"). A TODO in code is not a register.

Talking to product — their currency

- Not "the code is bad" but: "every checkout change costs +3 days and 2 of the last 5 incidents started there; 2 sprints of work removes both". Cost of delay, risk, and the roadmap item it unblocks.
- Offer options with trade-offs (do now / with feature X / accept and monitor), let them choose — then record the choice. Report the payoff after, so the next request is trusted.

RFC / design doc vs ADR

	RFC / design doc	ADR
purpose	propose, gather feedback before deciding	record a decision + its context
when	big or cross-team change	every architecturally significant choice
size	pages: goals, non-goals, alternatives, rollout, risks	1 page
lifecycle	draft → review → accepted / rejected	proposed → accepted → deprecated / superseded by ADR-n
lives	doc tool or repo, comment threads	repo, docs/adr/0012-title.md, beside the code

An accepted RFC usually *produces* one or more ADRs. ADRs are **immutable**: to change your mind, write a new ADR that supersedes the old one — the history of *why* is the value.

Remember

"Principal + interest; pay where interest is highest; say it in product terms; write down why."

Example — an ADR (Nygard's template)

```
# ADR 0012: Move iOS dependencies from CocoaPods to SPM
Status: Accepted, 2026-03-02. Supersedes ADR 0003.
Context: CocoaPods trunk stops accepting new podspecs on 2 Dec 2026, so pods get no new versions. We use 14 pods; 11 ship an SPM package. pod install adds ~4 min to every CI job and needs a Ruby toolchain.
Decision: Move all 14 to SPM this quarter, one PR per pod. The 3 without SPM: vendor the vendor's binary XCFramework (2) and replace the 3rd with ~200 lines.
Consequences:
+ one resolver + Package.resolved; no Ruby on CI
- ~2 weeks of work; 2 vendored SDKs updated by hand
```

Good ADR: **context** states the forces (facts, constraints, numbers), **decision** is active voice ("we will"), **consequences** lists the *bad* ones too. Also useful: options considered.

Build vs buy (or open source)

- Buy / adopt** what is not your differentiator and hard to do well (crash reporting, payments, maps, video). **Build** what *is* the product, or when cost grows with scale, or privacy, compliance, offline or latency rule the vendor out.
- Compare **total cost of ownership**: licence + integration + upkeep + upgrades + **exit cost** (the migration off it). **Cap the exit cost**: wrap the SDK behind *your* protocol (adapter / port) in one module; its types never leak into the domain.
- Scorecard** (1-5 × weight, table goes in the ADR): fit (3) · maintenance (3) · security + privacy (3) · size + transitive deps (2) · platform support — min OS, Swift 6, RN New Arch (2) · exit cost (2) · docs (1) · **licence = pass/fail gate**.

Traps

- ! "Let's rewrite it" as the debt plan — strangle it in shippable steps.
- ! Calling everything you dislike "debt": no recurring cost = no interest = no priority.
- ! A yearly "debt sprint" — debt returns; pay continuously, next to features.
- ! Editing or deleting an old ADR — supersede it; the old context is the point.
- ! Picking a library by GitHub stars and forgetting what leaving it will cost.